

**Name:** Bharath Arun Gandhimani

**Student ID:** 35501308

**Applied Session:** 7 | Teaching Associates: Michael Niemann & Ayden Zhou

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**Project Title:** From Roads to Classrooms: Educational Infrastructure Investment, Socioeconomic Equity, and Household Travel Behavior in Victoria (2023–2025)

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### **Aims:**

This visualisation communicates three connected findings to Victorian policymakers and transport planners:

- Education infrastructure investment broadly tracks where households are located across Victoria's LGAs
- Higher investment tiers correlate with greater car dependency, not active or public transport uptake
- Neighbourhood disadvantage — not investment level alone — is the stronger predictor of how households travel

The goal is to show where government spending and community transport need are aligned, and where they are not.

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### **Motivation:**

Victoria spends billions annually on regional infrastructure — but investment alone does not guarantee equity. If funding concentrates in already well-connected areas, or if economically disadvantaged households cannot access viable transport options, the gap between government intention and community outcome widens. This project combines VISTA 2024–25 travel survey data, the 2023–24 State Budget Infrastructure Investment Map, and ABS SEIFA 2021 indexes to assess whether infrastructure decisions are spatially and socially aligned with real household travel demand — and what that means for equity in daily mobility across Victoria.

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## IDEAS

01



Stacked Bar Chart

02



Grouped Bar Chart

03



Connected Dot Plot

04



Violin + Box Plot

05



Scatter + Trend Line

06



Lollipop Chart

07



Slope Chart

08



Diverging Bar Chart

09



Heatmap

10



Matrix / Table

11



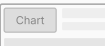
Small Multiples

12



Ranked Dot Plot

13



Scroll Narrative

14



Tab Dashboard

15



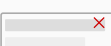
Stepper Layout

16



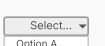
Bump Chart

17



Static Log Page

18



Dropdown Filter

19



Range Slider

20



Tooltip

21



Radio Button

22



Toggle

23



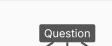
Linked Highlight

24



Headline Numbers

25



Question Decomp.

26



Data Coverage Gap Callout

## FILTER

#9 → Heatmap: poor at differentiating small % change across cells.

#10 → White-cell table: individual unit encoding becomes unreadable.

#2 → Toggle: interaction complexity without clear payoff

#17 → Static log page: needs interactivity to be meaningful.

#6 → Lollipop: redundant with bar + dot plot combination.

#8 → Diverging bar: realistic neutral midpoint may not exist in this data.

#13 → Scroll narrative: JavaScript dependency — excluded per course constraints.

#23 → Linked highlight: patterns matter more than individual row comparison.

## CATEGORISE

## Ideas that Set Context

#24, #25, #12

Orients the reader before analysis. Establishes scale, frames questions, shows what is most prominent.

## Ideas that Frame the Narrative

#25, #26

Tells the reader what data can and cannot show. #26 serves double duty — contextualises entry point and frames the whole narrative.

## Ideas that Show Comparisons

#1, #2, #3, #5

Lets the reader see differences between groups or across an ordered axis.

## Ideas that Show Distribution &amp; Variability

#4, #6

Reveals spread and shape behind summary stats. #11 can serve as companion across subgroups and as a layout device.

## Ideas that Give Reader Control

#18, #19, #20, #21

Interactive controls. Augment charts, not standalone. Risk: complexity without payoff for a policymaker audience.

## COMBINE &amp; REFINE

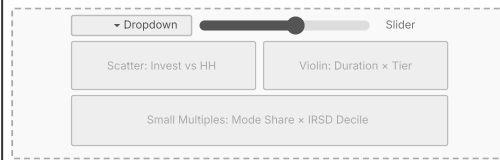
## Design A — Sequential Stepper

Fully author-driven. Fixed forward-only path. Each step: one chart (choropleth / lollipop / slope / grouped bar / scatter) + narrative text panel. Reader cannot skip ahead.



## Design B — Question-Answer Dashboard

Fully reader-driven. Three panels simultaneously visible using distinct encodings: scatter (bivariate), violin (distribution), small multiples (faceted comparison). Scoped filters per panel.



## Design C — Martini Glass

Guided scroll narrative (author-driven) through three headline-and-chart panels, then a state-change button unlocks an LGA Explorer. Author controls the evidence sequence; reader controls the depth.



## QUESTIONS

Q1 Does any design give a policymaker a clear sense of what action to take?

Design A ends at a clear policy implication. B and C leave the reader to draw their own conclusions, risking unresolved questions for a non-expert audience.

Q2 How much narrative text does a policymaker need alongside each chart?

Design A allocates explicit text panels per chart. B and C prioritise chart real estate over narration — risky for a policymaker who needs framing to act.

Q3 If a reader enters mid-way, can they understand without reading earlier sections?

Design B handles this best — each panel is self-contained. Design A breaks without sequential reading. C partially handles it via the persistent headline numbers in the title panel.

Q4 Does Design A's fixed sequence risk losing a reader interested in only one question?

Yes, but acceptable — policymakers need the full picture to act. Author-controlled narrative is appropriate for this audience.

Q5 Does Design B's simultaneous visibility create cognitive overload?

Yes. Showing all three questions at once without a guided entry overwhelms a non-expert reader. Works for analysts, not policymakers.

Q6 Does Design C's Martini Glass structure balance author-controlled narrative with reader-driven exploration, and is the two-state transition clear enough to prevent premature exploration?

Mostly yes. The narrative panels deliver the complete argument before the exploration zone unlocks, so readers who exit early still receive the full message. The main risk is that a persistent right column signals exploration before it is earned — the overlay copy and [D] button must be calibrated to preserve the author's intended reading order.

Q7 How do we communicate the structural data gaps without it looking like incomplete analysis?

Frame the 29-LGA coverage gap as a policy-relevant finding: VISTA does not survey regional Victoria, meaning policymakers currently lack travel behaviour evidence to evaluate investment in those areas. Present as a prominent callout, not a limitation footnote — the gap itself is part of the story.

## DVP PART 1 FEEDBACK — Revision Record

Michael Niemann identified that the three alternative designs were too similar — primarily layout variants rather than genuinely distinct design approaches. What structural changes were made for DVP Part 2?

The revised designs differ in three dimensions simultaneously. (1) **Narrative genre**: Design A is fully author-locked with forward-only stepper navigation; Design B is fully reader-controlled with simultaneous scoped panels; Design C uses a two-state Martini Glass — author-controlled in State A, reader-controlled in State B. (2) **Chart family**: Design A uses geographic and distribution encodings (choropleth, lollipop, slope, grouped bar, scatter); Design B uses bivariate, distributional, and faceted encodings (scatter, violin, small multiples); Design C uses strip, ranking, and paired-comparison encodings (dot strip, bump, connected dot). No chart type appears in more than one design. (3) **Interaction model**: Design A has no filters — locked forward navigation only; Design B has asymmetrically scoped cross-panel filters; Design C has no filters — only a state-change toggle between guided narrative and LGA personalisation. These differences operate at the level of spatial grammar, perceptual task, and interaction philosophy.

## INFORMATION

**Title:** Design A — Sequential Stepper | **Sheet No:** Sheet #2

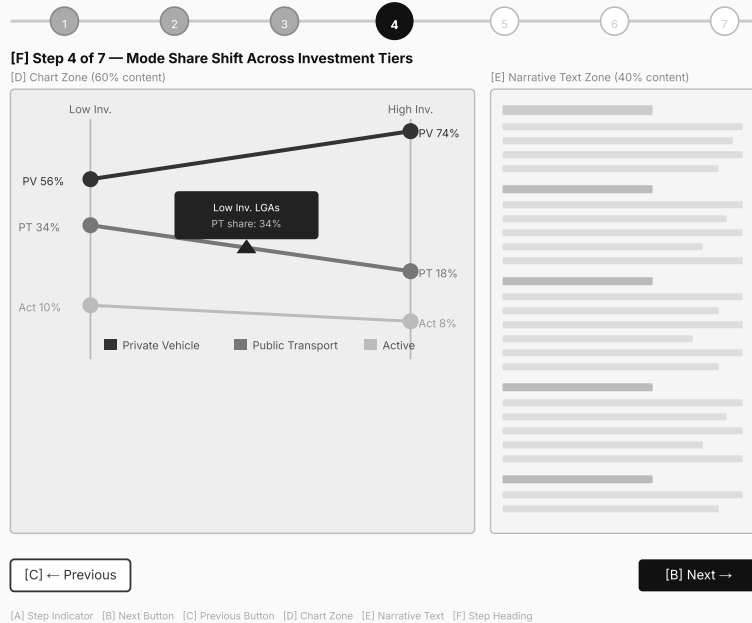
**Author:** Bharath Arun Gandhimani | **Date:** 12/05/2026

**Description:** Fully author-driven interactive narrative visualisation communicating findings about educational infrastructure investment, socioeconomic equity, and household travel behaviour in Victoria. Targeted at policymakers and transport planners. Reader progresses through a fixed 7-step sequence — one analytical layer per step.

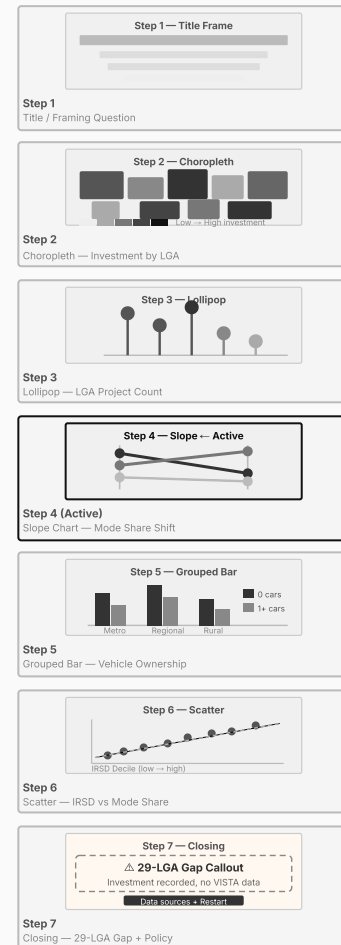
## DESIGN RATIONALE

The stepper enforces a deliberate reading order — investment geography is established first before socioeconomic or behavioural patterns are introduced. This prevents a policymaker from jumping to conclusions without the geographic context that frames the entire argument. Chart types (choropleth, slope chart, lollipop) are drawn from Sheet 1 Ideas #7, #6, and the geographic/map category respectively. No filters or dropdowns — narrative control stays with the author. Chart family (choropleth, lollipop, slope, grouped bar, scatter) is entirely distinct from Design B (scatter, violin, small multiples) and Design C (dot strip, bump, connected dot) — these three designs use non-overlapping visual encodings, satisfying the FDS requirement for genuine chart-level distinctness across Sheets 2-4.

## LAYOUT



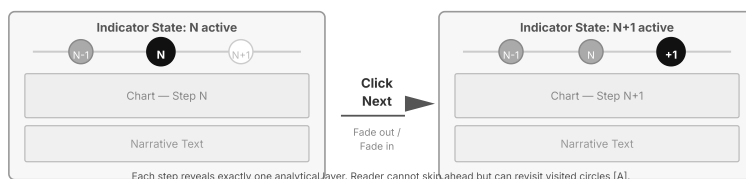
## OTHER STEPS



## OPERATIONS

|                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>[B] Click "Next →"</b><br>⇒<br>Content in [D] and [E] fades out. Next step's chart and narrative text fade in. Step indicator [A] advances — current circle fills solid grey (visited), next circle activates black. On the final step the Next button is replaced by a "Restart" option. | <b>[C] Click "← Previous"</b><br>⇒<br>Returns to the prior step. [A] updates to reflect the previous active step. [C] is hidden when the reader is on Step 1 — there is no "before" the opening frame.                                  | <b>[A] Click a visited step circle</b><br>⇒<br>Jumps directly to that completed step for review. Unvisited future circles remain greyed out and unclickable — the reader cannot skip ahead to content they have not yet reached.         |
| <b>[D] Hover on chart element</b><br>⇒<br>A tooltip appears showing exact values — percentage, count, LGA name. This is the only interactivity within a step. No filters, no dropdowns, no view changes are available.                                                                       | <b>[A] Step circle — state logic</b><br>⇒<br>Visited circles show filled grey. Active circle shows filled black. Unvisited future circles show white with light grey border and are unclickable. State maintained for the full session. | <b>Step 7 — Final step reached</b><br>⇒<br>The 29-LGA gap callout and policy implication text are revealed. Data source links and a "Restart" button appear. The [B] Next button is hidden. Reader has completed the full narrative arc. |

## FOCUS / PARTI — STEP TRANSITION MECHANISM



The closing step only makes sense once all prior evidence has been absorbed — this is what separates Design A from B and C. Chart types per step: Step 2 → Choropleth (geographic category, Sheet 1); Step 3 → Lollipop (Sheet 1 Idea #6); Step 4 → Slope Chart (Sheet 1 Idea #7).

## Step Circle States

- Visited — filled grey
- Active — filled black
- Locked — white, unclickable

## Label Reference

- [A] Step Indicator
- [B] Next Button
- [C] Previous Button
- [D] Chart Zone
- [E] Narrative Text
- [F] Step Heading

## DISCUSSION

## Pros

- Sees evidence in exact order as intended — no confusion about where to start.
- One chart, one explanation per step keeps things simple and focused for a non-technical policymaker audience.
- Opens with geographic framing (choropleth). Closes with what remains unknown (29-LGA gap). Clean narrative arc.
- Works well with non-technical audiences like policymakers who need framing before data.
- Easy to implement in R Shiny using a reactive step counter with conditional panel rendering.
- Chart types in this design — choropleth, lollipop, slope, grouped bar, scatter — address geographic and categorical comparisons. This is a different perceptual task family from Design B (distributional / bivariate) and Design C (strip / ranking / paired), ensuring genuine design-space breadth across Sheets 2-4 as required by the FDS methodology.

## Cons

- Cannot compare two charts side by side — reader must go back and forth.



INFORMATION

Title: Design B — Question-Answer Dashboard | Sheet No: Sheet #3

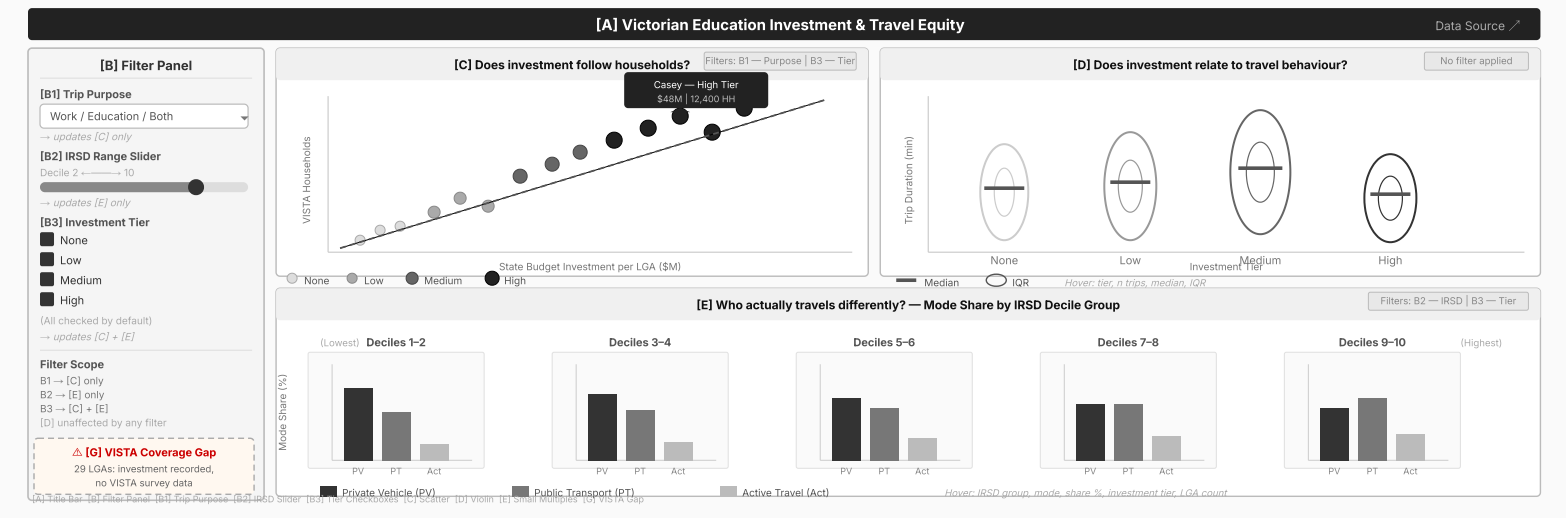
Author: Bharath Arun Gandhimani | Date: 13/05/2026

Description: Fully reader-driven dashboard communicating findings about educational infrastructure investment, socioeconomic equity, and household travel behaviour in Victoria to policymakers. Three chart panels are visible simultaneously — each poses an explicit question and answers it with a distinct encoding. A left-side filter panel scopes each filter to specific panels only.

DESIGN RATIONALE

Each panel uses a different encoding matched to its analytical task: scatter plot (bivariate investment-vs-household relationship), violin plots (distribution shape across investment tiers), and small multiples (faceted mode share by IRSD decile). Chart types drawn from Sheet 1 Ideas #5 (scatter), #4 (violin), #11 (small multiples). Filters are intentionally scoped — [B1] affects [C] only, [B2] affects [E] only, [B3] affects [C] and [E] — to prevent unintended cross-panel interference. [D] is deliberately unfiltered so the reader always has a stable distributional reference. No chart type in this design (scatter, violin, small multiples) appears in Design A or Design C — the three designs collectively cover geographic, distributional, bivariate, faceted, strip, ranking, and paired-comparison encodings across Sheets 2–4.

LAYOUT



INFORMATION

**Title:** Design C — Martini Glass | **Sheet No:** Sheet #4

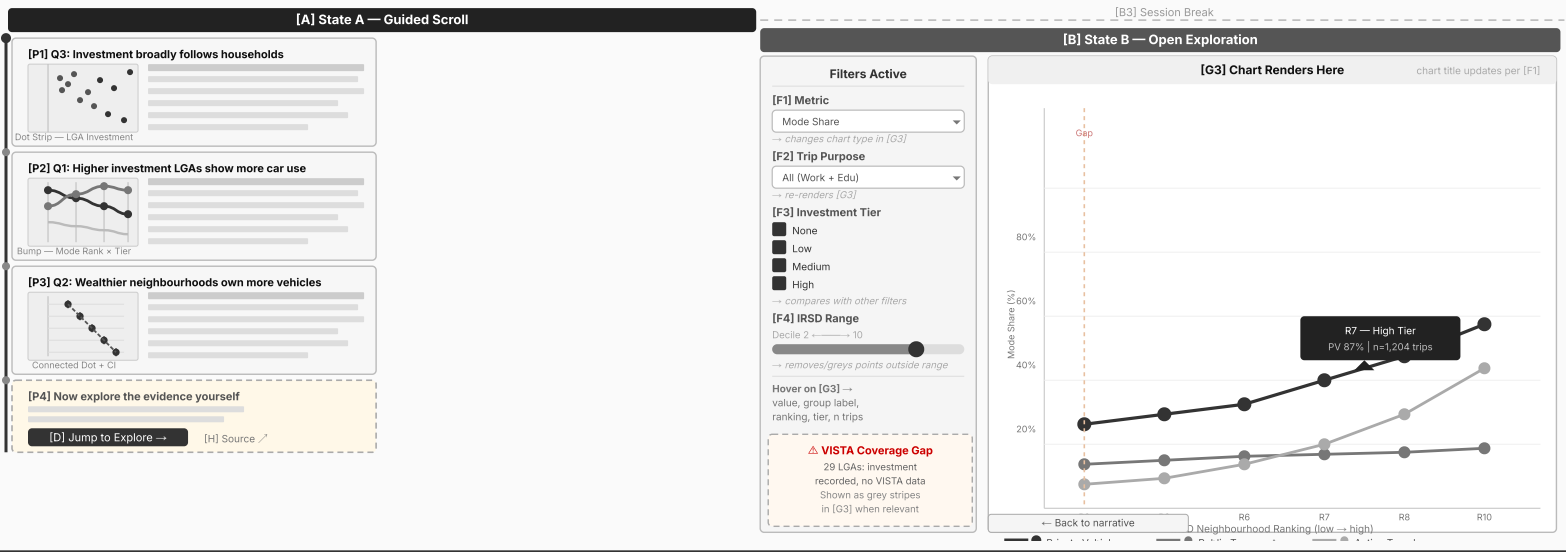
**Author:** Bharath Arun Gandhimani | **Date:** 13/05/2026

**Description:** Interactive narrative visualisation communicating findings about educational infrastructure investment, socioeconomic equity and household travel behaviour in Victoria to policymakers and transport planners. A two-state structure: State A delivers a guided scroll through four sequential narrative panels [P1–P4], each pairing a bold headline claim with a small embedded chart. A [D] button transitions the reader to State B, where filters [F1–F4] drive a live chart zone [G3] for open exploration.

DESIGN RATIONALE

The martini glass structure gives policymakers the narrative framing they need before exploration — author controls the story entry, reader controls the exit. Chart types are chosen to be fully distinct from Sheets 2 and 3 in both chart family AND analytical task — geographic/categorical (Sheet 2) vs distributional/bivariate (Sheet 3) vs strip/ranking/paired (Sheet 4): **dot strip plot** (investment dispersion across LGAs, Q3), **bump chart** (mode share rank shift across investment tiers, Q1), and **connected dot plot with 95% CI error bars** (vehicle ownership by income band, Q2). [F1] Metric dropdown switches the chart type in [G3] — a mode share metric renders a connected dot plot, a duration metric renders a violin+box plot, a vehicle ownership metric renders a connected dot plot with CI bars.

LAYOUT



OPERATIONS

[A] Scroll / natural read through [P1–P3]

→

Reader moves through three authored panels in sequence. Each panel is a fixed layout: chart thumbnail (left) + headline claim + narrative text (right). No interaction required — author controls the evidence order. [P4] bridge panel appears at the bottom of the scroll.

[D] Click "Jump to Explore"

→

Transitions from State A to State B. Left narrative column [P1–P3] is replaced by the filter panel [F1–F4]. Filters activate from greyed state. [G3] renders with default settings: F1 = Mode Share, F2 = All, F3 = All tiers checked, F4 = full IRSD range. A "← Back to narrative" link stays visible.

[F1] Change Metric dropdown

→

[G3] chart type changes: Mode Share → connected dot plot; Trip Duration → violin + box plot; Vehicle Ownership → connected dot plot with 95% CI error bars. Chart title in [G3] updates accordingly. All other active filters [F2–F4] remain applied.

[F2] Change Trip Purpose dropdown

→

[G3] re-renders for selected purpose subset (Work / Education / Both). Compares with [F1] and [F3] simultaneously. Chart title updates to reflect selected purpose.

[F3] Change Investment Tier checkboxes

→

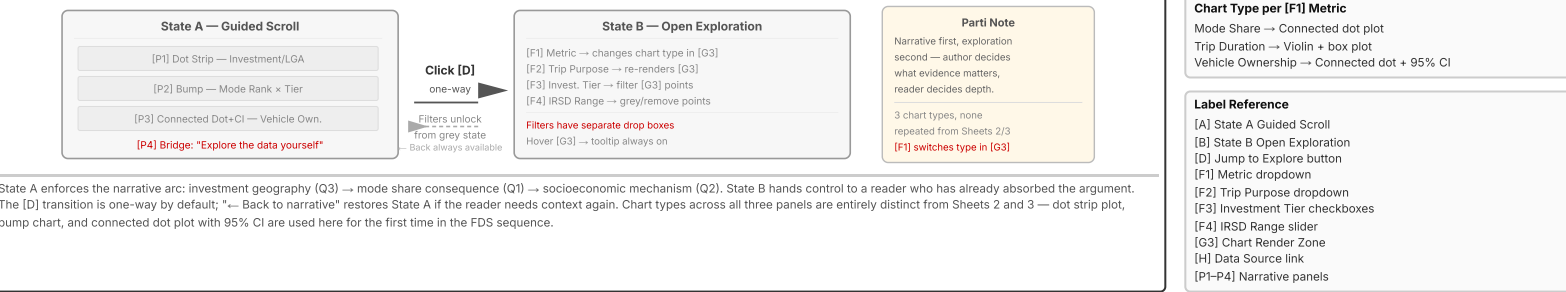
[G3] re-renders for selected tier(s) only. Unchecked tiers are removed from view. Chart title updates. Compounds with other active filters [F1] and [F2].

[F4] Drag IRSD Range slider

→

Data points outside selected IRSD decile range are removed or greyed out in [G3]. Compounds with other filters. Hover on any point in [G3] → tooltip: value, group label, ranking, tier, n trips. [H] data source link always accessible.

FOCUS / PARTI — TWO-STATE TRANSITION



INFORMATION

**Title:** Realization — Victoria Is Building Schools. But Not Transport. | **Sheet No:** Sheet #5

**Author:** Bharath Arun Gandhimani | **Date:** 17/05/2026

**Description:** Interactive narrative visualisation communicating findings about educational infrastructure investment, socioeconomic equity and household travel behaviour in Victoria to policymakers and transport planners. Martini glass structure: persistent two-column layout. Left column is a fixed three-part authored narrative scroll (State A). Right column activates as an LGA Explorer (State B) when reader signals readiness. No filters. Hover tooltips are the sole interactivity in the narrative zone.

DESIGN RATIONALE & LINEAGE

Synthesises across all prior sheets. Martini glass architecture from Sheet 4. Dot strip chart [P1] adapted from Sheet 3 [C] scatter. Connected line chart [P2] adapted from Sheet 4 [P2] slope chart — redesigned with a consistent investment tier x-axis for legibility. Connected dot plot [P3] from Sheet 4 [P3]. Right-zone filters [F1-F4] from Sheet 4 **removed** — filter-driven exploration places analytical burden on the reader inappropriate for a policymaker audience. Replaced by a single LGA dropdown that personalises an already-delivered finding to a specific suburb. 29-LGA gap callout from Sheet 1 Idea #26, promoted to a persistent right-zone element visible in both states.

